DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2019/2020

SECOND YEAR SUPPLEMENTARY EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CCS 2208: COMPUTER GRAPHICS

DATE: NOVEMBER 2019 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) Given a light source, describe three factors that affect the appearance of an object. **(3 marks)**

b) Suppose RGB raster system is to be designed using an 8 inch X 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer, how much storage (in kilobytes) do we need for frame buffer? **(3 marks)**

c) Distinguish between the following terms
   i. Scaling and Shearing **(2 marks)**
   ii. Antialiasing and pixel phasing **(2 marks)**
   iii. Window port and view port **(2 marks)**

d) The Phongs reflection model is a sum of three terms that each models a specific type of light reflection. Name and describe the three terms. **(3 marks)**

e) Discuss the architecture of a virtual reality system. Illustrate your answer. **(4 marks)**

f) The reflection along the line y=x is equivalent to the reflection about the X axis followed by counter clockwise rotation by θ degrees. Find the value of θ. **(4 marks)**

g) Identify and explain three types of colour models. **(3 marks)**

h) Suppose you have an image which you want to store in permanent storage, describe four factors you would consider when selecting a file format. **(4 marks)**

Question Two: (20 Marks)

a) Define affine transformation giving at least two properties. **(3 marks)**

b) Using homogeneous coordinates and composite matrices, rotate a triangle with vertices (10,20), (10,10), (20,10) about the origin by 30 degrees and then translate it by tx=5, ty=10. **(5 marks)**

c) Rasterize a line having end points (10,20) and (16,25) and plot the resultant pixels using Bresenham's algorithm. **(6 marks)**

d) Suppose you are given a scene to render through the graphics pipeline. Explain the Cartesian reference frames and procedure you would use to construct and display the scene. **(6 marks)**

Question Three: (20 Marks)

a) Briefly discuss the purpose of the following methods used in creating 3-D sceneries
   i. Parallel projection **(2 marks)**
   ii. Perspective projection **(2 marks)**

b) Given 3D geometric primitives explain how you would generate an image using the 3D rendering pipeline. **(6 marks)**

c) Describe two data compression methods giving one technique used in each compression category. **(3 marks)**

d) Write a simple OpenGL program for rendering on a black background a red square and a blue triangle. **(7 marks)**

Question Four: (20 Marks)

a) Identify and explain three advantages of LCDs. **(3 marks)**

b) Consider two raster systems with the resolutions of 1280 x 720 and 2048 x 1536.
   i. Describe two ways of refreshing a display screen. **(2 marks)**
   ii. How many pixels could be accessed per second in each of these systems by a display controller with a refresh rate of 120Hz? **(2 marks)**
   iii. What is the access time per pixel in each system? **(2 marks)**

c) Film animators use a variety of methods for depicting and emphasizing motion sequences. Discuss four traditional animation techniques giving examples. **(8 marks)**

d) How is colour represented in computer graphics and how does this relate to the Human Visual System. **(3 marks)**